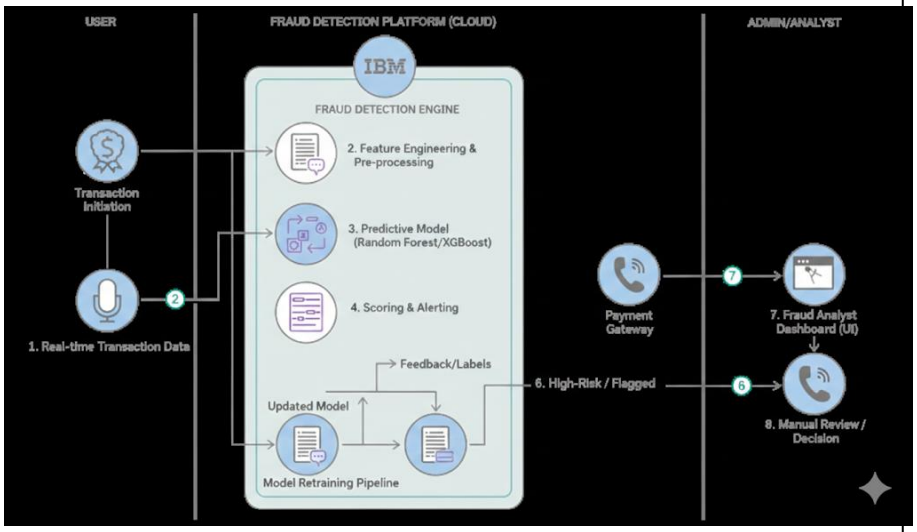
Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	31 January 3035
Team ID	LTVIP2026TMIDS76805
Project Name	Online Payments Fraud Detection using Machine Learning
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Online Payments Fraud Detection using Machine Learning



Guidelines:

Include all the processes (As an application logic / Technology Block)
Provide infrastructural demarcation (Local / Cloud)
Indicate external interfaces (third party API's etc.)
Indicate Data Storage components / services
Indicate interface to machine learning models (if applicable)

Table-1 : Components & TechnoloS.No	Component	Description	Technology
1	User Interface	Web interface where users enter transaction details and view fraud prediction results	HTML, CSS, JavaScript, Bootstrap
2	Application Logic-1	Handles user input validation and request processing	Python, Flask
3	Application Logic-2	Fraud detection processing logic	Scikit-learn, Pandas, NumPy
4	Application Logic-3	Fraud probability calculation and result display	Flask Backend
5	Database	Stores user details and transaction records	MySQL
6	Cloud Database	Cloud-based storage for scalability (optional)	Firebase / MySQL Cloud
7	File Storage	Stores trained machine learning model (.pkl file)	Local File System
8	External API-1	Email alert service for suspicious transactions	SMTP / Gmail API
9	External API-2	OTP verification service	Twilio API
10	Machine Learning Model	Predicts whether a transaction is Fraud or Safe	Random Forest / XGBoost
11	Infrastructure (Server / Cloud)	Deployment of application on local or cloud server	Local Server / AWS / Render

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used	Technology of Opensource framework
2.	Security Implementations	List all the security / access controls implemented, use of firewalls etc.	e.g. SHA-256, Encryptions, IAM Controls, OWASP etc.
3.	Scalable Architecture	Justify the scalability of architecture (3 – tier, Micro-services)	Technology used
4.	Availability	Justify the availability of application (e.g. use of load balancers, distributed servers etc.)	Technology used
5.	Performance	Design consideration for the performance of the application (number of requests per sec, use of Cache, use of CDN's) etc.	Technology used

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology Used
1	Open-Source Frameworks	Open-source tools used for development	Flask, Scikit-learn, Bootstrap
2	Security Implementations	Secure authentication, encrypted communication, session control	SHA-256, HTTPS, Flask-Login
3	Scalable Architecture	3-Tier architecture allowing scaling of backend and database	Flask + MySQL + Cloud Hosting
4	Availability	Application accessible 24/7 using cloud infrastructure	AWS / Render Cloud
5	Performance	Real-time fraud prediction with optimized ML model and minimal response time	Optimized ML Model, Caching